

A stable coordinate that is not a coordinate in four variables

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Abstract

We construct an explicit degree-five polynomial over $\mathbf{C}$ that is not a coordinate in four variables but becomes one after adjoining a single variable. This gives a counterexample to the stable coordinate conjecture in four variables. Every fiber is isomorphic to affine three-space, yet its embedding in affine four-space is not rectifiable. Thus the example also disproves the Abhyankar–Sathaye embedding conjecture in ambient dimension four.

1 Introduction

Let $k = \mathbf{C}$. An element $f \in k[x_1, \dots, x_n]$ is a *coordinate* if there are elements $f_2, \dots, f_n$ such that

$$k[x_1, \dots, x_n] = k[f, f_2, \dots, f_n]$$

as a polynomial ring. Equivalently, a polynomial automorphism sends f to x_1 . It is a *one-stable coordinate* if it is a coordinate in $k[x_1, \dots, x_n, w]$, where w is independent. The stable coordinate conjecture asks whether an element that becomes a coordinate after adjoining finitely many variables was already a coordinate. The one-variable formulation appears in [12, Conjecture 3]; successive applications give the formulation with finitely many added variables. Over $\mathbf{C}$, the answer is affirmative in two and three variables by [12, Proposition 1.1(c)]. The case of one original variable is elementary. Thus four variables are the first possible dimension for a counterexample, and one added variable already suffices.

Theorem 1.1. *In $R = \mathbf{C}[x_1, x_2, x_3, x_4]$, put*

$$Q = x_2^2 - x_4^2 + x_1x_3, \quad f = x_1 - 2Q(Q(x_2 + x_4) + x_1x_4). \quad (1.1)$$

There is a $\mathbf{C}$ -algebra automorphism of $R[w]$ sending f to x_1 , but there is no such automorphism of R .

The example also has a geometric consequence. A closed embedding $\mathbb{A}^{n-1} \hookrightarrow \mathbb{A}^n$ is *rectifiable* if an automorphism of the ambient affine space takes its image to a coordinate hyperplane. The Abhyankar–Sathaye embedding conjecture asserts, in characteristic zero, that a polynomial g with $k[x_1, \dots, x_n]/(g) \simeq k^{[n-1]}$ is a coordinate; see [2, p. 8429]. Here $k^{[r]}$ denotes a polynomial ring in r variables. Corollary 1.2 gives a negative answer to this conjecture in ambient dimension four. In fact, every fiber is affine three-space and its inclusion in $\mathbb{A}^4$ is nonrectifiable.

Corollary 1.2. *For every $\lambda \in \mathbf{C}$, the hypersurface $f^{-1}(\lambda) \subset \mathbb{A}^4$ is isomorphic to $\mathbb{A}^3$ and its embedding in $\mathbb{A}^4$ is not rectifiable.*

Proof. Proposition 2.3 identifies each fiber with $\mathbb{A}^3$; in particular, $(f - \lambda)$ is its prime defining ideal. If an ambient automorphism takes $f^{-1}(\lambda)$ to a coordinate hyperplane, it takes its defining principal ideal to the ideal of a coordinate. The two generators therefore differ by a nonzero constant. This would make $f - \lambda$, and hence f , a coordinate, contrary to Theorem 1.1. $\square$

History and related methods

The coordinate question retains the embedding in affine space: it asks whether a given function can be made into a coordinate projection by an ambient automorphism. Cancellation asks instead whether an affine variety whose product with an affine line is an affine space must itself be an affine space. Makar-Limanov, van Rossum, Shpilrain, and Yu established an implication from the stable coordinate conjecture to cancellation in [10, Theorem 1.2]. In our example the ambient variety is already $\mathbb{A}^4$, and the obstruction concerns its distinguished polynomial.

The low-dimensional proof in [12, proof of Proposition 1.1(c)] uses curve cancellation and the Abhyankar–Moh embedding theorem in two variables. In three variables it combines affine-surface cancellation, using work of Fujita and Miyanishi–Sugie, with Kaliman’s theorem that a polynomial on $\mathbb{C}^3$ with general fiber $\mathbb{C}^2$ is a coordinate [6, Main Theorem].

The four-variable Vénéreau polynomials have been an important testing ground for these coordinate questions. Lewis proved that the second Vénéreau polynomial is a coordinate [9]. Blanc and Poloni developed a description in terms of affine bundles and bivariables, revisited that result, and proved one-stability for a broader family of related polynomials [1, Proposition 2.3]. These results illustrate the distinction between exhibiting polynomial coordinates after stabilization and proving that a polynomial cannot occur in any original coordinate system. For the polynomial in Theorem 1.1, the first problem is settled by explicit substitutions; the second requires a derivation obstruction.

The construction and the obstruction

The proof uses two presentations of the same four-variable polynomial ring. We first write it as a hypersurface $A = P/(H)$ in a five-variable polynomial ring P , with a distinguished element p . A determinant-preserving matrix transformation makes H linear in one new coordinate. It identifies A with R and carries p to the polynomial in (1.1). A second construction, using the exponential of a locally nilpotent derivation, shows that p belongs to polynomial coordinates on $A[w]$. These explicit changes of coordinates occupy Section 2.

To show that p is not a coordinate, we put an integer-indexed filtration on A . Its associated graded ring has the form

$$G = B[\tau, I/\tau] \subset B[\tau, \tau^{-1}], \quad B = k[x, y, z, u]/(xy - z(z + 1)),$$

where I is an explicitly defined two-generated ideal of B and τ is the initial form of p . The notation $B[\tau, I/\tau]$ means the subalgebra generated by B , τ , and all elements h/τ with $h \in I$. If p were a coordinate, differentiation in a suitable other coordinate would induce a nonzero locally nilpotent derivation E on B whose kernel contains a nonzero element of I . Section 3 proves this implication, including the care needed for negative filtration degrees.

The last two sections exclude that invariant. Section 4 passes from $\text{Spec } B$ to the complement of the zero section in a line bundle. In its coordinate ring

$$C = k[a, d, b, c, u]/(ac - bd - 1),$$

the ideal I becomes principal, generated by $v = a^3b + a^2u^2 - d^2$. The additive action defined by E lifts to this bundle and commutes with fiber scaling. A nonzero invariant in I would force the lifted derivation to annihilate v . In Section 5, a second grading reduces this to a derivation fixing a, d, u . Such a derivation has a coefficient that would simultaneously have to be a polynomial in a, d, u and have negative fiber weight. This is impossible.

The graded-ring and line-bundle strategy adapts the construction in [11, Sections 2–6]. That paper studies an affine-space cancellation example. Here the quadratic choice in the defining equation has two different uses: it permits the determinant-preserving change that makes A polynomial, and it makes the highest auxiliary part of v factor as $(au - d)(au + d)$. The latter factorization gives the final rigidity argument without a polynomial abc estimate. The passage to leading derivations follows the filtered methods of Kaliman and Makar-Limanov [7, Sections 3–5], and the lift of an additive action uses the line-bundle normalization argument of Brion [3, Lemma 2.9]. Every construction and obstruction needed for Theorem 1.1 is proved below.

2 Polynomial presentation and stabilization

We construct the counterexample through a hypersurface presentation of a four-variable polynomial ring. This presentation makes stabilization explicit and will also supply the filtration used to obstruct coordinateness. It adapts the construction of [11, Section 2], with quadratic terms that permit the polynomial change of coordinates below.

Let

$$\begin{aligned} P &= k[p, s, u, F, J], & x &= s^2 - u^2 + pF, \\ H &= x^2F - (1 + 2sx)J - pJ^2 - u, & A &= P/(H). \end{aligned} \tag{2.1}$$

The five generators of P are independent. We use the same letters for their images in A ; the distinguished element is $p \in A$.

2.1 A polynomial change of coordinates

Proposition 2.1. *There are polynomial coordinates p', s', F', J on A such that*

$$x = (s')^2 - J^2 + p'F', \quad p = p' - 2x(x(s' + J) + p'J). \tag{2.2}$$

Thus, under the identification $(p', s', F', J) = (x_1, x_2, x_3, x_4)$, the element p is the polynomial f in Theorem 1.1.

Proof. The relation H becomes linear after a change of coordinates preserving x . To exhibit that change, write

$$M = \begin{pmatrix} F & s - u \\ s + u & -p \end{pmatrix}, \quad e = \begin{pmatrix} x \\ -J \end{pmatrix}, \quad n = \begin{pmatrix} J \\ x \end{pmatrix}.$$

Then $x = -\det M$, $e^t n = 0$, and

$$H = e^t M e - J - u.$$

Keep J fixed and define F', s', u', p' by

$$M' = (I_2 - 2ne^t)M = \begin{pmatrix} F' & s' - u' \\ s' + u' & -p' \end{pmatrix}. \tag{2.3}$$

Since $(ne^\dagger)^2 = 0$ and $\det(I_2 - 2ne^\dagger) = 1$, we have $-\det M' = x$. Consequently x and J , and hence e and n , are unchanged by the substitution. Its inverse is the polynomial substitution

$$M = (I_2 + 2ne^\dagger)M', \quad x = -\det M',$$

so p', s', u', F', J form a coordinate system on P .

Taking half the difference of the off-diagonal entries in Equation (2.3) gives

$$u' = u - e^\dagger M e, \quad H = -J - u'.$$

Eliminating u' therefore identifies A with $k[p', s', F', J]$. In this quotient $u' = -J$, so the determinant identity gives the first formula in Equation (2.2). The lower-right entry of the inverse matrix identity gives

$$p = p' - 2x(x(s' - u') + p'J),$$

which becomes the second formula after substituting $u' = -J$. $\square$

The expression for p in Equation (2.2) takes the values 0 and 1 at $(p', s', F', J) = (0, 0, 0, 0)$ and $(1, 0, 0, 0)$, respectively. In particular, p is a nonzero nonunit of A .

2.2 One-variable stabilization

We next change the defining equation after adjoining one variable, while keeping p fixed. For this purpose set

$$y = s + x(x + u^2), \quad z = sx + pJ. \quad (2.4)$$

Expansion gives the identity

$$xy - z(z + 1) = p(H + u). \quad (2.5)$$

Moreover,

$$P[p^{-1}] = k[p, p^{-1}][x, y, z, u], \quad (2.6)$$

with inverse coordinate formulas

$$s = y - x(x + u^2), \quad F = \frac{x - s^2 + u^2}{p}, \quad J = \frac{z - sx}{p}. \quad (2.7)$$

Proposition 2.2. *For an independent variable w , the ring $A[w]$ is a polynomial ring in five variables with p as one of its coordinates.*

Proof. A derivation δ of a k -algebra is *locally nilpotent* if, for every element g , some iterate $\delta^r(g)$ is zero. We use such a derivation to replace the equation $H = 0$ by $H + pw = 0$.

In the coordinates of Equation (2.6), take

$$\Delta = -p \frac{\partial}{\partial u}, \quad \Delta(p) = \Delta(x) = \Delta(y) = \Delta(z) = 0.$$

This locally nilpotent derivation of $P[p^{-1}]$ preserves P , because Equation (2.7) yields

$$\begin{aligned} \Delta(p) &= 0, & \Delta(u) &= -p, & \Delta(s) &= 2pxu, \\ \Delta(F) &= -4sxu - 2u, & \Delta(J) &= -2x^2u. \end{aligned} \quad (2.8)$$

Its restriction to P is therefore locally nilpotent as well. Applying Δ to Equation (2.5) gives $\Delta(H) = p$. Extend Δ to $P[w]$ by $\Delta(w) = 0$. The exponential

$$\exp(w\Delta)(g) = \sum_{r \geq 0} \frac{w^r \Delta^r(g)}{r!}$$

is a polynomial automorphism: each sum is finite, and its inverse is $\exp(-w\Delta)$. It fixes p and sends H to $H + pw$, inducing a $k[p]$ -algebra isomorphism

$$A[w] \simeq P[w]/(H + pw). \quad (2.9)$$

It remains to eliminate one variable from the equation on the right. Put $x_0 = s^2 - u^2$ and introduce

$$\begin{aligned} L &= x_0^2 F - (1 + 2sx_0)J, \\ N &= (1 - 2sx_0)F + 4s^2 J. \end{aligned} \quad (2.10)$$

The coefficient matrix of this substitution has determinant $4s^2 x_0^2 + (1 + 2sx_0)(1 - 2sx_0) = 1$; explicitly,

$$F = 4s^2 L + (1 + 2sx_0)N, \quad J = -(1 - 2sx_0)L + x_0^2 N. \quad (2.11)$$

Thus p, s, u, L, N are polynomial coordinates on P . Expanding $x = x_0 + pF$ in H gives

$$H = L - u + pQ_0, \quad Q_0 = 2x_0 F^2 + pF^3 - 2sFJ - J^2. \quad (2.12)$$

Since $Q_0 \in P$, the substitution $w' = w + Q_0$ is another polynomial change of coordinates, and

$$P[w]/(H + pw) = k[p, s, u, L, N, w']/(L - u + pw') \simeq k[p, s, u, N, w'].$$

Together with Equation (2.9), this proves the assertion with p fixed throughout. Composing with the identification in Proposition 2.1 and relabeling the target variables so that p maps to x_1 gives an automorphism of $R[w]$ sending f to x_1 . $\square$

The two steps explicitly realize the exponential shift of [4, Lemma 3.3] and the case of the transfer construction in [5, Theorem 7] in which the perturbation pQ_0 is divisible by p .

2.3 The fibers

The same coordinate computations describe every fiber of p . This will give a geometric consequence once noncoordinateness has been proved.

Proposition 2.3. *For every $\lambda \in k$, the quotient $A/(p - \lambda)$ is a polynomial ring in three variables over k .*

Proof. At $\lambda = 0$, Equations (2.10)–(2.12) give

$$A/(p) = k[s, u, L, N]/(L - u) = k[s, u, N].$$

For $\lambda \neq 0$, Equations (2.6) and (2.5) give

$$A/(p - \lambda) = k[x, y, z, u]/(xy - z(z + 1) - \lambda u) \simeq k[x, y, z],$$

by elimination of u . $\square$

Propositions 2.1 and 2.2 establish the polynomial presentation and the stable-coordinate assertion of Theorem 1.1. The remaining task is to show that p cannot be a coordinate of A itself.

3 A graded obstruction to coordinateness

We now turn to the assertion that p is not a coordinate of A . The first step is to pass to a weighted degeneration of A . If p were a coordinate, differentiation in another coordinate would then produce a nonzero locally nilpotent derivation of a simpler quadric ring, with a nonzero invariant in an explicit ideal. The rest of the proof will exclude that possibility. The use of leading derivations follows the filtration methods of [7, Sections 3–5]; the particular quadric construction is adapted from [11, Sections 3–5].

3.1 The associated graded ring

Give the generators of $P = k[p, s, u, F, J]$ the weights

$$\begin{array}{c|ccccc} & p & s & u & F & J \\ \hline \text{weight} & -1 & 0 & 0 & 1 & 1. \end{array} \quad (3.1)$$

For $d \in \mathbb{Z}$, let $\mathcal{F}_d P$ be the span of the monomials of weight at most d , and give $A = P/(H)$ the quotient filtration $\mathcal{F}_d A = \text{im}(\mathcal{F}_d P \rightarrow A)$. These are increasing multiplicative filtrations, indexed by all integers. Their negative indices are essential: p has weight -1 .

Proposition 3.1. *The filtration on A is exhaustive and separated. Its associated graded ring is the domain*

$$G := \text{gr } A \simeq k[p, s, u, F, J]/(H_*), \quad H_* = x^2 F - (1 + 2sx)J - pJ^2, \quad x = s^2 - u^2 + pF, \quad (3.2)$$

with the grading induced by (3.1). The filtration degrees of p, s, u, F, J in A are respectively $-1, 0, 0, 1, 1$.

Proof. Exhaustiveness follows because every element of A has a polynomial representative. For $r > 0$, every monomial of weight at most $-r$ is divisible by p^r , so

$$\mathcal{F}_{-r} A \subseteq p^r A.$$

By Proposition 2.1, A is a polynomial ring, and Equation (2.2) shows that p is a nonzero nonunit. Therefore $\bigcap_{r>0} p^r A = 0$: in a unique factorization domain, a nonzero element cannot be divisible by arbitrarily high powers of an irreducible factor of p . This proves separation. In particular, every nonzero $a \in A$ has a well-defined degree

$$\deg a = \min\{d \in \mathbb{Z} : a \in \mathcal{F}_d A\}$$

and a nonzero initial form in $\mathcal{F}_d A / \mathcal{F}_{d-1} A$.

The element x is homogeneous of weight zero in P , and $H = H_* - u$ has highest weight part H_* , of weight one. For every nonzero $q \in P$, the highest weight part of qH is $(\text{in } q)H_*$: the polynomial ring is a graded domain, so the product of the highest parts does not vanish. Consequently the ideal of initial relations is exactly (H_*) . For completeness, the natural graded map $P \rightarrow \text{gr } A$ sends a homogeneous polynomial of weight d to its class in $\mathcal{F}_d A / \mathcal{F}_{d-1} A$, which may at this stage be zero. This map is surjective. A homogeneous polynomial of weight d lies in its kernel exactly when its image in A has a representative of weight at most $d - 1$; subtracting that representative gives a relation with the original polynomial as its highest part. Thus the kernel is the ideal of initial relations, proving (3.2).

To prove that this quotient is a domain, use the polynomial coordinates x, y, z, u of $P[p^{-1}]$ from the preceding section. The identity $xy - z(z + 1) = p(H + u)$ becomes

$$pH_* = xy - z(z + 1).$$

The polynomial on the right is irreducible over $k[p, p^{-1}, x, z, u]$: as a polynomial in y , it is primitive, because x and $z(z+1)$ are coprime, and it has degree one over the fraction field. Thus H_* is irreducible after inverting p . It is not divisible by p before localization, since its reduction modulo p is

$$(s^2 - u^2)^2 F - (1 + 2s(s^2 - u^2))J \neq 0.$$

Any factorization of H_* in P has a factor that becomes a unit after inverting p , hence is a scalar times a power of p . Since $p \nmid H_*$, this factor is already a unit in P . Thus H_* is irreducible in P , and hence prime by unique factorization. Finally, H_* is not associated to any of the five variables, so their classes in G are nonzero. These classes are the initial forms of the corresponding elements of A , proving the asserted degrees. $\square$

3.2 A quadric and its distinguished ideal

From now on write τ for the class of p in G , to distinguish the graded ring from A . Consider the quadric ring

$$B = k[x, y, z, u]/(xy - z(z+1)). \quad (3.3)$$

The formulas

$$x = s^2 - u^2 + \tau F, \quad y = s + x(x + u^2), \quad z = sx + \tau J$$

define elements of degree zero in G . They satisfy the relation in (3.3); after inverting τ , the inverse coordinate formulas from the preceding section give an isomorphism

$$G[\tau^{-1}] = B[\tau, \tau^{-1}], \quad \deg B = 0, \quad \deg \tau = -1. \quad (3.4)$$

In particular, the induced map $B \rightarrow G$ is injective: its composition with localization is the inclusion of B in this Laurent polynomial ring. We henceforth regard B as a subring of G .

Define elements and an ideal of B by

$$\begin{aligned} s_B &= y - x(x + u^2), \\ f_0 &= x - s_B^2 + u^2, \quad g_0 = z - s_B x, \quad I = (f_0, g_0). \end{aligned} \quad (3.5)$$

In (3.4) we have $s = s_B$, $F = f_0/\tau$, and $J = g_0/\tau$. These identities give the more precise description

$$G = B[\tau, I/\tau] = B[\tau, f_0/\tau, g_0/\tau] \subset B[\tau, \tau^{-1}]. \quad (3.6)$$

This is the algebraic construction of an affine modification, with divisor $\tau = 0$ and center (τ, I) in $B[\tau]$; see [8, Definition 1.1 and Proposition 1.1(b)]. We will only need its explicit homogeneous pieces.

Lemma 3.2. *For each integer $\ell > 0$, the degree- ℓ piece of G is*

$$G_\ell = \tau^{-\ell} I^\ell.$$

Proof. A monomial in the generators in (3.6), with coefficient in B , has the form

$$b \tau^r (f_0/\tau)^i (g_0/\tau)^j, \quad r, i, j \geq 0.$$

It has degree ℓ exactly when $i + j - r = \ell$. In that case it equals $\tau^{-\ell} b f_0^i g_0^j$, whose numerator belongs to $I^{\ell+r} \subseteq I^\ell$. Since the presentation is homogeneous, such monomials span G_ℓ . Conversely, every element of I^ℓ is a B -linear combination of the products $f_0^i g_0^j$ with $i + j = \ell$, and dividing these products by τ^ℓ gives elements of G_ℓ . $\square$

3.3 What a coordinate would force

The ideal I records the positive part of the grading. We now show why a hypothetical coordinate system containing p supplies a nonzero locally nilpotent derivation of B fixing a nonzero element of I .

Proposition 3.3. *If p is a coordinate of A , then there exist a nonzero locally nilpotent derivation E of the ring B in (3.3) and an element $0 \neq h \in I$ such that $E(h) = 0$.*

Proof. Suppose $A = k[p, q_1, q_2, q_3]$ for a polynomial coordinate system. At least one q_i has strictly positive filtration degree. Indeed, if all three had degree at most zero, then so would every polynomial in p, q_1, q_2, q_3 , contrary to $\deg F = 1$. Relabel so that $\deg q_1 > 0$, and take

$$D = \frac{\partial}{\partial q_2}.$$

This is a nonzero locally nilpotent derivation of A satisfying $D(p) = D(q_1) = 0$.

We give the details of its passage to G , because the filtration has both positive and negative indices. Write $\omega(t)$ for the assigned weight of a generator $t \in \{p, s, u, F, J\}$, and put

$$m = \max_{D(t) \neq 0} (\deg D(t) - \omega(t)). \quad (3.7)$$

This maximum is defined: a nonzero derivation cannot kill all algebra generators. The product rule shows that a monomial of weight at most d has derivative in $\mathcal{F}_{d+m}A$. Every element of $\mathcal{F}_dA$ has a representative that is a sum of such monomials, by the definition of the quotient filtration. Thus

$$D(\mathcal{F}_dA) \subseteq \mathcal{F}_{d+m}A \quad (d \in \mathbb{Z}).$$

It follows that D induces a homogeneous derivation D_0 of G of degree m . For $a \in \mathcal{F}_dA$, write $[a]_d$ for its class in $\mathcal{F}_dA/\mathcal{F}_{d-1}A$. Explicitly,

$$D_0([a]_d) = [D(a)]_{d+m}, \quad D_0^r([a]_d) = [D^r(a)]_{d+rm} \quad (r \geq 0). \quad (3.8)$$

The first formula is well-defined because D also sends $\mathcal{F}_{d-1}A$ into $\mathcal{F}_{d+m-1}A$; the second follows by induction. It proves that D_0 is locally nilpotent, first on homogeneous elements and then on finite sums of them. Moreover, equality is attained in (3.7) for some generator t , whose initial form has degree $\omega(t)$ by Proposition 3.1. Its image under D_0 is the nonzero initial form of $D(t)$, so $D_0 \neq 0$. Finally, (3.8) shows that D_0 kills both τ and the nonzero initial form η of q_1 . The latter is homogeneous of degree $\ell = \deg q_1 > 0$.

Because $D_0(\tau) = 0$, extending D_0 to $G[\tau^{-1}] = B[\tau, \tau^{-1}]$ preserves local nilpotence. Homogeneity implies that there is a derivation $E : B \rightarrow B$ with

$$D_0(b) = \tau^{-m}E(b) \quad (b \in B). \quad (3.9)$$

Indeed, every degree- m element of the Laurent ring is uniquely τ^{-m} times an element of B . The derivation E is nonzero: otherwise D_0 would kill both B and τ , hence the whole Laurent ring and its subring G . It is locally nilpotent because

$$E^r(b) = \tau^{rm}D_0^r(b) \quad (b \in B, r \geq 0).$$

Here, as in (3.8), the formula follows by induction, using that D_0 fixes τ .

By Lemma 3.2, write $\eta = \tau^{-\ell}h$ with $0 \neq h \in I^\ell$. Equations (3.9) and $D_0(\eta) = 0$ give

$$0 = D_0(\eta) = \tau^{-m-\ell}E(h).$$

Thus $E(h) = 0$, and $I^\ell \subseteq I$ because $\ell > 0$. □

Proposition 3.3 has removed the original coordinate system from the problem. It remains to show that no nonzero locally nilpotent derivation of B can have a nonzero invariant in I . We will make this ideal principal after pulling it back to the complement of the zero section in a line bundle, where a second grading gives the contradiction.

4 Lifting the obstruction to a line bundle

Proposition 3.3 reduces noncoordinateness to excluding a nonzero locally nilpotent derivation E of B with a nonzero invariant in $I = (f_0, g_0)$. We pull I back to the complement of the zero section in a line bundle over $\text{Spec } B$; on this complement the ideal becomes principal. Every additive action on the base lifts linearly to the line bundle and preserves this complement. Consequently, an invariant in I will force the lifted action to fix one specific polynomial. This use of the quadric and its line bundle adapts the construction of [11, Sections 4–6]; all the lifting and principalization arguments needed here are proved below.

Put $Y = \text{Spec } B$ and define

$$C = k[a, d, b, c, u]/(ac - bd - 1). \quad (4.1)$$

The assignment

$$x = ab, \quad y = dc, \quad z = db, \quad u = u \quad (4.2)$$

defines a homomorphism $B \rightarrow C$, since $xy = abdc = (db)(ac) = z(z + 1)$. The ring C is a domain: the polynomial $ac - bd - 1$, viewed as a primitive linear polynomial in c over $k[a, d, b, u]$, is irreducible. We give C the *fiber weights*

$$\text{wt}(a, d, b, c, u) = (1, 1, -1, -1, 0). \quad (4.3)$$

The relation is homogeneous of weight zero, and the image of B has weight zero. These weights describe the action of $\mathbb{G}_m$ that scales the fibers of the following bundle. Geometrically this is the familiar quotient of SL_2 by a diagonal torus, times the affine u -line; see [2, p. 8431]. We use explicit charts throughout.

Lemma 4.1. *The map $B \rightarrow C$ is injective. The morphism $\text{Spec } C \rightarrow Y$ is the complement of the zero section in a line bundle $\mathcal{L} \rightarrow Y$. It is trivial over the open cover*

$$U_0 = D(z + 1), \quad U_1 = D(z)$$

of Y , with respective invertible fiber coordinates a and d .

Proof. Over U_0 , the identity $ac = z + 1$ makes a invertible. The formulas

$$c = \frac{z + 1}{a}, \quad b = \frac{x}{a}, \quad d = \frac{ay}{z + 1} \quad (4.4)$$

give an isomorphism $C[(z + 1)^{-1}] = B[(z + 1)^{-1}][a, a^{-1}]$. In particular the localized map from B is injective, and so is the original map because B is a domain. Over U_1 , the corresponding formulas are

$$b = \frac{z}{d}, \quad c = \frac{y}{d}, \quad a = \frac{xd}{z}, \quad (4.5)$$

and give $C[z^{-1}] = B[z^{-1}][d, d^{-1}]$. The opens cover Y because z and $z + 1$ generate the unit ideal. On their overlap the fiber coordinates satisfy

$$d = \frac{y}{z + 1} a, \quad (4.6)$$

where $y/(z + 1)$ is a unit with inverse x/z . Gluing the trivial line bundles by this transition gives $\mathcal{L}$; removing their zero sections gives precisely the two charts of $\text{Spec } C$ above. Both fiber coordinates have weight one, so the scaling agrees with (4.3). $\square$

The two generators of I acquire a common factor in C . Define

$$v = a^2(x + u^2) - d^2 = a^3b + a^2u^2 - d^2 \in C. \quad (4.7)$$

Lemma 4.2. *The extended ideal IC is the principal ideal (v) .*

Proof. In $C[a^{-1}]$, put $\xi = d/a$ and $j = x + u^2 - \xi^2$. The determinant relation and (4.2) give

$$z = \xi x, \quad y = \xi + \xi^2 x, \quad s_B = y - x(x + u^2) = \xi - xj.$$

Consequently

$$f_0 = (1 + 2\xi x - x^2 j)j, \quad g_0 = x^2 j, \quad v = a^2 j. \quad (4.8)$$

The two coefficients of j generate the unit ideal, as the identity

$$(1 - 2\xi x)(1 + 2\xi x - x^2 j) + x^2(4\xi^2 + j - 2\xi x j) = 1$$

shows. Thus $IC[a^{-1}] = (j) = (v)$.

It remains to check primes of C containing a . Modulo such a prime, $bd = -1$, while $x = 0$ and $z = -1$. Hence $g_0 = z - s_B x$ has nonzero residue -1 , and v has nonzero residue $-d^2$. Both are units in the corresponding local ring. The ideals IC and (v) therefore agree at every prime, which proves the assertion. $\square$

Figure 1 records the geometric meaning of this ideal equality: the inverse image of $V(I)$ is cut out by the single function v .

$$\begin{array}{ccc} V(v) & \hookrightarrow & \text{Spec } C \\ \downarrow & & \downarrow \\ V(I) & \hookrightarrow & Y = \text{Spec } B \end{array}$$

Figure 1: A Cartesian square of closed subschemes. Pulling I to the complement of the zero section of $\mathcal{L}$ gives the principal ideal (v) . In particular, every $h \in I$ pulls back to a multiple of v .

We next lift additive actions from Y to $\mathcal{L}$. The elementary reason that a lift exists is that adjoining an affine-line parameter does not change line bundles on Y .

Lemma 4.3. *Every line bundle on $\mathbb{A}^1 \times Y$ is isomorphic to the pullback of its restriction to $\{0\} \times Y$.*

Proof. The two charts of Lemma 4.1 have coordinate rings

$$\begin{aligned} B[(z+1)^{-1}] &= k[x, t, u, (xt+1)^{-1}], & t &= \frac{y}{z+1}, \quad z = xt, \\ B[z^{-1}] &= k[x, t, u, (xt-1)^{-1}], & t &= \frac{y}{z}, \quad z = xt-1. \end{aligned}$$

Here t denotes a separate coordinate on each chart. Each ring, and its polynomial extension by the affine-line coordinate T , is a noetherian unique factorization domain. A line bundle on the spectrum of such a ring is trivial: its divisor class is represented by a Cartier divisor, and unique factorization makes every divisor principal.

Let $\mathcal{M}$ be a line bundle on $\mathbb{A}^1 \times Y$. Choose trivializations on $\mathbb{A}^1 \times U_0$ and $\mathbb{A}^1 \times U_1$. Its transition function belongs to $B[z^{-1}, (z+1)^{-1}][T]^*$. A polynomial over a domain is a unit only when it is a constant unit, so this transition function is independent of T . Its specialization at $T = 0$ is therefore the same transition function. The chosen trivializations identify $\mathcal{M}$ with the pullback of its restriction at zero. $\square$

The lifting argument has a useful general form. Its normalization step is the standard one for linearizing a line bundle under an additive action; compare [3, Lemma 2.9].

Lemma 4.4. *Let X be an affine integral k -variety such that every line bundle on $\mathbb{A}^1 \times X$ is the pullback of its restriction at zero. Every $\mathbb{G}_a$ -action on X lifts to any line bundle on X by maps linear on fibers, commuting with fiber scaling.*

Proof. Write $R = k[X]$, let $\mathcal{M}$ be the line bundle, and let $\alpha : \mathbb{A}^1 \times X \rightarrow X$ be the action. If $\pi : \mathbb{A}^1 \times X \rightarrow X$ is projection, then $\alpha^*\mathcal{M}$ and $\pi^*\mathcal{M}$ restrict to $\mathcal{M}$ at parameter zero. The hypothesis therefore supplies an isomorphism

$$\eta : \pi^*\mathcal{M} \rightarrow \alpha^*\mathcal{M}.$$

At zero, η is an automorphism of $\mathcal{M}$, hence multiplication by a unit of R . Multiplying η by the inverse pullback of that unit normalizes it to the identity at zero.

The isomorphism η gives a fiberwise linear lift of α . To check the group law, compare, over $\mathbb{A}^2 \times X$, the map on fibers obtained by applying parameters S and then T with the map obtained by applying $T + S$. Both are global isomorphisms of the same source and target line bundles, with fiber maps from $\mathcal{M}_x$ to $\mathcal{M}_{\alpha(T+S,x)}$. Their ratio is an automorphism of the source line bundle. Its endomorphism sheaf is canonically $\mathcal{O}$, so this automorphism is multiplication by a global unit in $R[T, S]$. Since R is a domain, this unit is independent of T, S . At $(T, S) = (0, 0)$ it equals one by the normalization of η . Thus the group law holds. The resulting action is linear on fibers, so it commutes with fiber scaling and preserves the complement of the zero section. $\square$

We will also use the following elementary property of locally nilpotent derivations; see [7, Proposition 2.1(3)].

Lemma 4.5. *Let R be a domain over a field of characteristic zero and let δ be a locally nilpotent derivation of R . If $r, s \in R$ are nonzero and $\delta(rs) = 0$, then $\delta(r) = \delta(s) = 0$.*

Proof. The exponential $\exp(T\delta)$ takes values in $R[T]$. If rs is fixed, then

$$\exp(T\delta)(r) \exp(T\delta)(s) = rs.$$

Degrees in T add under multiplication of nonzero polynomials over a domain. Both factors on the left therefore have degree zero, and their coefficients of T give the assertion. $\square$

Proposition 4.6. *Let E be a nonzero locally nilpotent derivation of B such that $I \cap \ker E \neq 0$. There is a nonzero locally nilpotent derivation $\tilde{E}$ of C that extends E , preserves fiber weights, and satisfies $\tilde{E}(v) = 0$.*

Proof. The exponential of E defines an additive-group action on Y . By Lemmas 4.3 and 4.4, it lifts to $\mathcal{L}$ by fiberwise linear maps. Restrict to the complement of the zero section, namely $\text{Spec } C$. Differentiating this action at the identity gives a derivation $\tilde{E}$ of C . The action law expresses its pullback on each function as $\sum_{r \geq 0} T^r \tilde{E}^r / r!$; because the action is algebraic, the sum is finite for each function. Hence $\tilde{E}$ is locally nilpotent. It extends E and is nonzero, and commutation with fiber scaling says exactly that it preserves the weights (4.3).

Finally choose $0 \neq h \in I \cap \ker E$. The inclusion $B \subset C$ and Lemma 4.2 give a factorization $h = vq$ in C , with both factors nonzero. Since $\tilde{E}(h) = 0$, Lemma 4.5 yields $\tilde{E}(v) = 0$. $\square$

5 Rigidity and the coordinate obstruction

Proposition 4.6 converts the hypothetical invariant in I into a locally nilpotent derivation of C that fixes v and preserves fiber weights. We now rule out such a derivation. An auxiliary grading isolates the difference of squares $a^2u^2 - d^2$ inside v . Its two factors force three generators to be invariant, leaving a coefficient whose required fiber weight is impossible.

Proposition 5.1. *There is no nonzero locally nilpotent derivation of $C = k[a, d, b, c, u]/(ac - bd - 1)$ that preserves the fiber weights (4.3) and annihilates $v = a^3b + a^2u^2 - d^2$.*

Proof. Suppose that $\tilde{E}$ is such a derivation. Give C the auxiliary grading in the second row of the following table:

	a	d	b	c	u
fiber weight	1	1	-1	-1	0
auxiliary degree	0	1	-1	0	1

(5.1)

The two gradings are compatible because $ac - bd - 1$ is homogeneous of degree zero for each. The auxiliary degrees of the two parts of v are

$$\deg_{\text{aux}}(a^3b) = -1, \quad \deg_{\text{aux}}(a^2u^2 - d^2) = 2.$$

Decompose $\tilde{E}$ into its homogeneous components for auxiliary degree shift, and let E' be the nonzero component with largest shift, say m . There are only finitely many components: a derivation is determined by its values on the five homogeneous generators, and each value is a finite sum of homogeneous terms. Each component is a derivation by the product rule. Moreover, E' is locally nilpotent. Indeed, for homogeneous $r \in C$ of auxiliary degree e , the component of auxiliary degree $e + nm$ in $\tilde{E}^n(r)$ is $(E')^n(r)$. For sufficiently large n the former expression vanishes, so the latter vanishes as well. Homogeneous elements span C , proving local nilpotence on every element. Since the gradings in (5.1) are compatible, E' still preserves fiber weights.

Taking the component of degree $m + 2$ in $\tilde{E}(v) = 0$ gives

$$E'(a^2u^2 - d^2) = 0.$$

The factorization

$$a^2u^2 - d^2 = (au - d)(au + d)$$

and Lemma 4.5 imply that E' annihilates both $au - d$ and $au + d$. Thus $E'(d) = E'(au) = 0$, and another application of that lemma gives

$$E'(a) = E'(d) = E'(u) = 0. \tag{5.2}$$

All the factors used here are nonzero; for example, after inverting a one has

$$C[a^{-1}] = k[a, a^{-1}, d, u][b], \quad c = \frac{1 + bd}{a}, \tag{5.3}$$

where their nonvanishing is immediate.

It remains to control the possible values of E' on b, c . Differentiating $ac - bd = 1$ and using (5.2) yields $aE'(c) = dE'(b)$. Define

$$q = cE'(b) - bE'(c) \in C.$$

Multiplication by a and d , respectively, then gives

$$E'(b) = aq, \quad E'(c) = dq. \tag{5.4}$$

The element q is nonzero, since otherwise E' would kill every generator. We claim that q belongs to the polynomial subring $k[a, d, u]$.

The elements a, d, u are algebraically independent by (5.3). Put $K = k(a, d, u)$ and localize C at the nonzero elements of $k[a, d, u]$. Because these elements lie in $\ker E'$, the localized derivation remains locally nilpotent. The resulting algebra is $K[b]$, with $c = (1 + bd)/a$. A locally nilpotent K -derivation of $K[b]$ sends b into K : if its value $P(b)$ had degree at least one, iterating $P(b) d/db$ on b would never give zero. More explicitly, differentiation of a nonconstant polynomial multiplies its nonzero leading coefficient by its positive degree, and multiplication by P then leaves the result nonconstant. Equation (5.4) therefore gives $q \in K$.

To compute $C \cap K$ inside the fraction field of C , use both coordinate charts:

$$C[a^{-1}] = k[a, a^{-1}, d, u][b], \quad C[d^{-1}] = k[a, d, d^{-1}, u][c].$$

An element of the first polynomial ring lying in K has no b terms, and an element of the second lying in K has no c terms. Hence

$$C \cap K \subset k[a, a^{-1}, d, u] \cap k[a, d, d^{-1}, u] = k[a, d, u].$$

The last equality follows from unique factorization in $k[a, d, u]$: a reduced denominator that is both a power of a and a power of d is a unit. The reverse inclusion is immediate, proving the claim.

Finally, preservation of fiber weights makes $E'(b)$ have weight -1 . By (5.4), the nonzero polynomial q has weight -2 . This is impossible in $k[a, d, u]$, whose generators have weights $1, 1, 0$. The contradiction proves the proposition. $\square$

Proof of Theorem 1.1. By Proposition 2.1, the algebra A is a polynomial ring in four variables and its element p becomes the polynomial f stated in the theorem. Proposition 2.2 makes p a coordinate of $A[w]$. If p were a coordinate of A , Proposition 3.3 would produce a nonzero locally nilpotent derivation E of B with $I \cap \ker E \neq 0$. Proposition 4.6 would then supply a derivation forbidden by Proposition 5.1. Thus p is not a coordinate of A . Transporting these two properties through the polynomial presentation proves the assertion for f . $\square$

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